

Calibrating a Frontier Task Tier for Architecture-Sensitive LLM Benchmarking

Status Report II — baseline results, full-size probe, and the frozen frontier-v2 tier

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Abstract

The pre-registered study compares monolithic, agentic (Executor+Verifier), and swarm (3-peer majority vote) LLM execution architectures under pinned model, prompts, and decoding. The v1 baseline (225 runs, frozen 2026-07-01) saturated: the pinned 14B model solves the classic benchmark items nearly perfectly, leaving architectures nothing to differentiate on. This report documents the construction and calibration of a *frontier tier* — tasks selected so that monolithic single-pass accuracy lands in the 0.4–0.7 band where iteration and voting can matter. A 330-run full-size probe (DeepSeek-R1-distill 32B) exposed both failure directions: MBPP code items saturated (100%) while closed-book MuSiQue floored (6%), and no architecture rescued the floor. A three-round calibration campaign (350 monolithic runs at the pinned 14B) re-sourced math to AIME 2025 (42%), code to BigCodeBench-Hard (48%), and multihop to MuSiQue *with* supporting passages (67%); the resulting 36-item tier is frozen. Two methodological findings fell out: per-item accuracy is near-binary under temperature-0 decoding (forcing the calibration band to the domain aggregate), and 47% of frontier failures are output-budget exhaustion rather than wrong reasoning. Two empirical surprises: the 14B distill *outperforms* its 32B sibling on the same MATH-500 slice (88% vs. 65%), and on floored knowledge tasks agentic/swarm wrappers scored *zero* where monolithic scraped 17%. The confirmatory tier $\times$ architecture sweep now moves to university HPC.

1 Context

Design, hypotheses (task-dependent superiority, “candidate D”), conditions, and metrics are frozen in `PRE_REGISTRATION.md` (2026-07-01 07:31 EDT); every deviation since is in its Amendment Log. Architecture is the only variable: one pinned model (DeepSeek-R1 14B distill, Q4_K_M, digest recorded), frozen prompts, temperature 0.0, fixed seed, $n_{ctx} = 8192$, max output 6144 tokens, $N = 5$ trials per cell. The harness grades binarily and automatically (exact numeric / unit tests / normalized string), writes one crash-safe JSONL row per run, and is idempotent.

2 v1 baseline (frozen 2026-07-01)

225 runs: 15 items (5 GSM8K / 5 HumanEval / 5 HotpotQA) $\times$ 3 architectures $\times$ $N=5$, local cell (Apple M5 Max, LM Studio/Metal).

Table 1: v1 baseline, all 15 items (mean $\pm$ std across runs).

Architecture	Acc.	Latency (s)	Tokens	tok/s
Monolithic	93%	15.2 $\pm$ 17.5	832 $\pm$ 850	44.9
Agentic	100%	32.5 $\pm$ 32.0	1961 $\pm$ 1662	44.5
Swarm	93%	32.9 $\pm$ 33.1	2649 $\pm$ 2247	63.0

The agentic verifier fixed every monolithic error (notably the one item all single-pass runs missed), at $\sim 2.4\times$ token cost; the swarm’s parallel peers raised throughput (63 tok/s) but its majority vote failed exactly where peers agreed on the same wrong answer. The problem is the 93–100% ceiling itself: with at most one hard item, the design cannot observe the *task-dependent* effects it pre-registered. Hence the frontier tier: a second, harder task class kept objectively auto-graded, selected by a pre-declared empirical rule — monolithic single-pass accuracy in $[0.4, 0.7]$ — the band where self-correction and voting have room to act.

3 Full-size probe: three regimes

Before calibrating at the pinned 14B, the 22 externally-sourced candidates (MATH-500 $L\geq 3$ / MBPP-sanitized / closed-book MuSiQue) were probed at full size: DeepSeek-R1-distill-**32B**, all three architectures, $N=5$ — 330 runs (~ 14 h). The probe exposed both failure directions at once (Table 2).

Table 2: 32B probe: accuracy by domain $\times$ architecture.

Domain	Mono	Agentic	Swarm	Verdict
Math (MATH-500)	70%	63%	63%	in band
Code (MBPP)	100%	100%	100%	ceiling
Multihop (closed-book)	17%	0%	0%	floor

Two observations survive any later analysis. First, *architecture cannot substitute for knowledge*: closed-book MuSiQue asks for facts the model does not hold, and iterating or voting over ignorance produced strictly *worse* results than a single pass (0/60 vs. 5/30) — the verifier “corrects” toward confident hallucinations, and peers vote on shared ones. Second,

on the one in-band domain (math), monolithic *beat* both wrappers at 32B (70% vs. 63%) — an early, exploratory contradiction of the hypothesis’s direction that the confirmatory 14B sweep will test properly. Mean cost ordering matched pre-registration sub-prediction 1: 1610 / 2706 / 2884 tokens per run (mono/agentive/swarm).

4 Calibration campaign and freeze

Calibration ran monolithic-only, $N=5$, against the *pinned* 14B (the model under test). Three rounds, 350 runs total (Table 3).

Table 3: Domain aggregates per calibration round (pinned 14B). Bold = in band, frozen.

Round	Domain source	Agg.	Action
1	MATH-500 L3-5 mix	88%	re-source harder
1	BigCodeBench (stdlib)	89%	re-source harder
1	MuSiQue + context	67%	freeze
2	MATH-500 L5-only	80%	re-source harder
2	BigCodeBench-Hard	48%	freeze
3	AIME 2025	42%	freeze

Two amendments shaped the rules (both logged before the affected runs):

Per-item binarity → **domain-aggregate band**. Under temperature 0.0 with a fixed seed, the five trials of an item are near-identical: observed per-item accuracies were 5/5 or 0/5 in 66 of 68 item-round combinations. A per-item 0.4-0.7 band is therefore unsatisfiable as written; the band was restated at the domain aggregate, with whole-domain re-sourcing from pre-declared harder tiers (MATH-500 → L5 → AIME 2025; BCB → BCB-Hard) and an explicit prohibition on item-level outcome selection.

Provided context for multihop. MuSiQue is calibrated upstream as reading comprehension; serving it closed-book turns it into a parametric-recall lottery (the 32B floor). v2 supplies the supporting passages plus distractors within a fixed context budget, making the domain reasoning-bound. It calibrated to 67% immediately.

The frozen tier (`tasks/frontier_v2_manifest.json`, 2026-07-09): 12 AIME 2025 math items (42%), 12 BigCodeBench-Hard code items (48%), 12 MuSiQue-with-context multihop items (67%); tier aggregate 52%. Every domain contains genuinely solved and genuinely unsolved items — the property the tier×architecture analysis needs. AIME 2025 additionally post-dates the pinned model’s training, removing the data-contamination concern the MATH-500 items carry.

5 Failure taxonomy at the frontier

Of the 86 incorrect rows in the frozen tier’s calibration, 46 are reasoning errors but **40 (47%) are output-budget exhaustions**: the model consumed all 6144 output tokens on chain-of-thought and was truncated before emitting any final answer. (The same mechanism forced the 2048→6144

amendment during the v1 pilot; at frontier difficulty it recurs at 6144.) This matters for interpretation: part of what “too hard for the pinned model” *means* under a fixed budget is “cannot finish reasoning in time.” It is also an architecture-relevant mechanism — whether a verifier retry or a differently-seeded peer can land an answer inside the budget where a single pass cannot is precisely a tier×architecture question.

6 Findings so far

- (1) **Baseline saturation is real and model-size-robust**: classic items (GSM8K/HumanEval/MBPP-class) are ceiling-level for both 14B and 32B; architecture comparisons on such suites measure noise.
- (2) **Capability is not monotonic in scale per-domain**: the 14B distill scored 88% on the same MATH-500 slice where the 32B scored 65%, while the 32B was far stronger on closed-book recall (its 6% floor still beat the 14B’s expected zero). Difficulty bands are model-relative; tiers must be calibrated per model.
- (3) **Architecture cannot fix missing knowledge**, and can amplify its absence (0% agentive/swarm vs. 17% mono on the floored domain).
- (4) **Budget exhaustion is a first-class failure mode** at frontier difficulty (47% of failures), distinct from wrong reasoning and reportable separately.
- (5) (Exploratory) **monolithic led both wrappers on in-band math at 32B**; whether this holds at the pinned 14B is exactly what the confirmatory sweep tests.

7 Threats to validity

Single model family (DeepSeek-R1 distills) and single environment so far (Metal/LM Studio; the pre-registered cloud cell is pending). Domain-aggregate calibration keeps selection mechanical but leaves per-domain aggregates at different band positions (42 / 48 / 67%), so cross-domain accuracy comparisons remain descriptive. AIME 2025 avoids contamination, but BigCodeBench-Hard and MuSiQue pre-date the model’s cutoff. $N=5$ under near-deterministic decoding characterizes serving jitter, not sampling variance. The 32B architecture observations use uncalibrated v1-candidate items and are exploratory only.

8 Next steps

- (1) Confirmatory sweep on university HPC: frozen tier × 3 architectures × $N=5$ at the pinned 14B (monolithic rows seeded from calibration), per `docs/HPC_RUNBOOK.md`;
- (2) exploratory scaling legs (32B, Qwen3.6-35B-A3B) at $N=2$;
- (3) LLM-as-judge secondary pass (Llama-3.2-3B);
- (4) the pre-registered Azure cloud cell;
- (5) future work: a tool-equipped v2 study arm where GAIA-style tasks become applicable — rejected for the current tier because it benchmarks tool orchestration, which the no-tool design deliberately excludes.

*All raw rows, calibration evidence, amendments, and scripts are committed; every figure and table in this report is reproducible from **results/** at the repository head.*